SPECATHON • IOT THEME

MISSION SANJEEVANI

AIoT-Enabled Emergency Medical Response Network

Smart Connectivity. Intelligent Response. Saving Lives.

SENSE **CONNECT** **ANALYZE** **RESPOND**



Government Polytechnic Madhubani

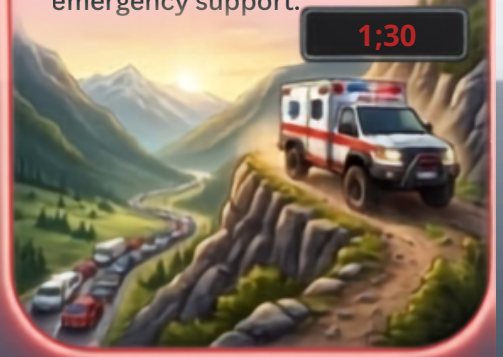
01 • THE WHY

THE GOLDEN-HOUR GAP

In emergencies, the challenge is not only medical—it is connectivity, information and time.

DELAYED ACCESS

Remote terrain, traffic and distance can slow emergency support.



FRAGMENTED DATA

Patient and scene information may not reach responders quickly.



CONNECTIVITY GAPS

Cloud-only systems may struggle when network coverage is weak.



SLOW COORDINATION

Multiple teams need a common, connected emergency picture.



THE QUESTION Can IoT connect sensing, communication, intelligence and physical response into one emergency ecosystem?

02 • OUR ANSWER

MISSION SANJEEVANI IS AN IOT SYSTEM

The drone is the mobility layer. The connected IoT ecosystem is the core innovation.

01 SENSE

IoT sensors



02 CONNECT

LoRaWAN + 4G/SG



03 THINK

Edge AI



04 DECIDE

Emergency control



05 ACT

Drone response



06 SUPPORT

Medical aid

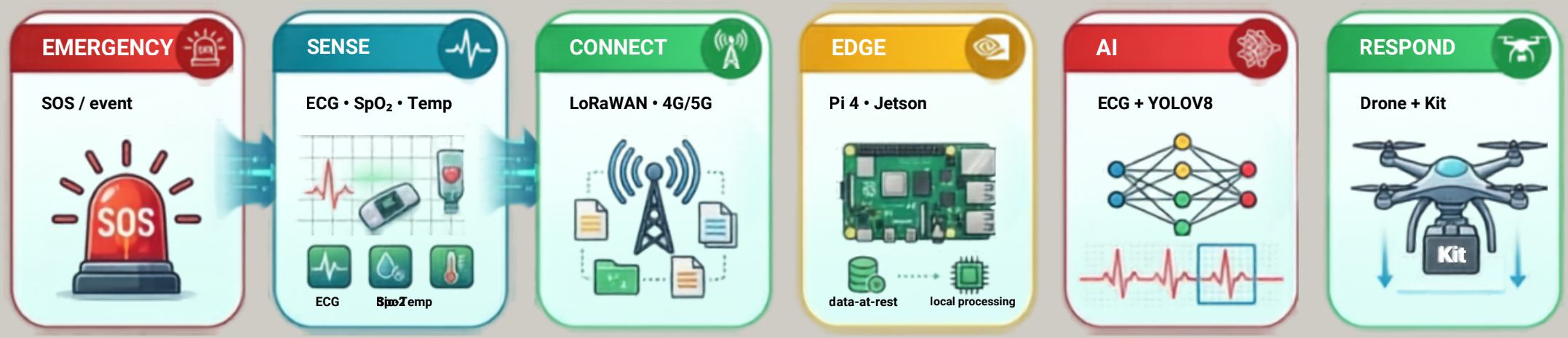


IoT → connects the physical world to the digital emergency response.

03 • SYSTEM ARCHITECTURE

END-TO-END IoT + AIOT FLOW

A resilient pipeline from real-world sensing to real-world action.



The system is designed to keep critical intelligence close to the edge while using hybrid connectivity for resilient communication.

THE IoT STACK

A practical stack built around sensing, connectivity, edge intelligence and response.

SENSING

ECG • SpO_2 • Temperature
Capture real-world emergency data

CONTROL

ESP32 • Raspberry Pi 4
Local device coordination

CONNECTIVITY

LoRaWAN • 4G/5G
Long-range + hybrid communication

EDGE AI

Jetson Nano • Edge AI
Local processing & decision support

VISION

YOLOv8
Landing-zone detection & navigation support

ACTION

Drone • Medical Payload
Translate digital response into physical aid

FROM SOS TO EMERGENCY SUPPORT

The connected loop is designed around speed, resilience and actionable information.

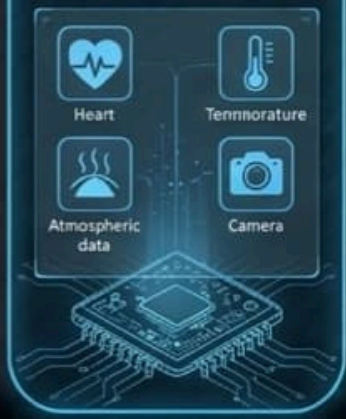
01 DETECT

SOS / emergency event is received.



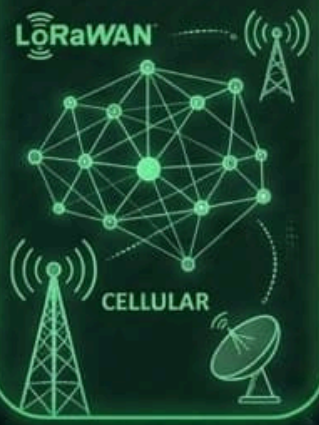
02 SENSE

IoT devices collect available vital information.



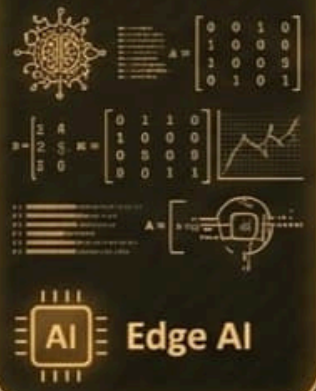
03 CONNECT

Data moves through LoRaWAN or cellular networks.



04 ANALYZE

Edge AI processes critical information locally.



05 RESPOND

Mission control coordinates the response.



06 DELIVER

Drone transports essential emergency medical supplies.



SENSE → CONNECT → ANALYZE → RESPOND → DELIVER

WHY THIS CAN STAND OUT

Mission Sanjeevani combines IoT with AI, edge computing and autonomous mobility.

CONNECTED

IoT links sensors, responders, drones and healthcare systems.



RESILIENT

Hybrid communication reduces dependence on a single network.



EDGE-ENABLED

Local processing supports operation in low-connectivity areas.



AI-READY

AI modules are being implemented and tested within the architecture.



AUTONOMOUS

YOLOv8 vision is planned for landing-zone recognition and navigation support.



SCALABLE

Future solar-hybrid stations can create distributed rural response nodes.

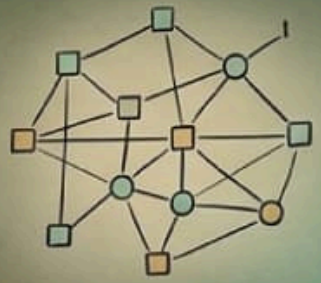


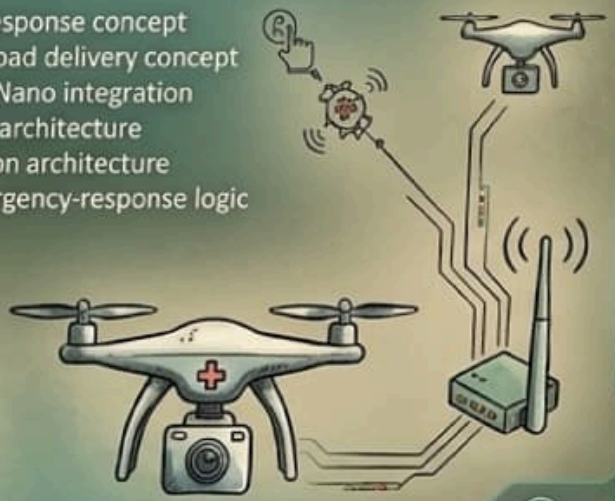
PROTOTYPE TODAY ⚙ AIoT TOMORROW

We clearly separate what is currently demonstrated from what is under development.

CURRENT PROTOTYPE / INTEGRATION

- IoT-oriented emergency response concept
- Drone-based medical payload delivery concept
- Raspberry PI 4 and Jetson Nano integration
- LoRaWAN communication architecture
- Edge AI and computer-vision architecture
- System workflow and emergency-response logic



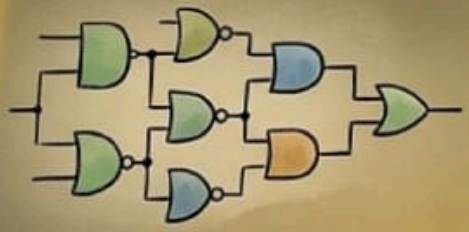


AIoT

AI MODULES UNDER IMPLEMENTATION & TESTING

- AI-based ECG analysis and decision support
- YOLOv8-based landing-zone detection
- Edge AI optimization for offline operation
- Autonomous navigation and landing improvements
- AI-assisted emergency assessment







AIoT

Our approach: demonstrate the connected architecture now → validate AI modules step-by-step → scale toward a complete AIoT system.

08 • FUTURE SCOPE

FROM IoT PROTOTYPE TO AIoT EMERGENCY NETWORK

A roadmap for making Mission Sanjeevani smarter, more autonomous and more scalable.

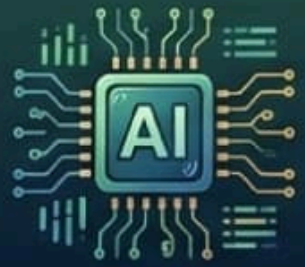
NOW

IoT prototype + connected architecture



NEXT

AI modules under implementation & testing



THEN

Edge AI + advanced ECG analytics



SCALE

Multi-drone fleet + smart routing



EXPAND

Solar-hybrid rural charging stations



SENSE — ANALYZE — RESPOND

VISION | A distributed rural emergency network where connected devices sense, AI analyzes, and autonomous systems respond.

WHY IT MATTERS

Technology with a measurable human purpose.

TIME

Reduce response delays by bringing early support closer to the emergency.



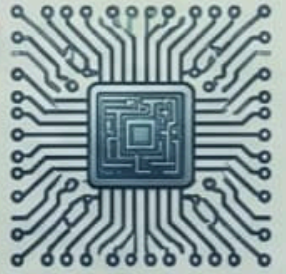
ACCESS

Extend emergency support to difficult-to-reach communities.



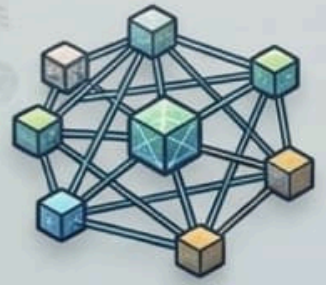
RESILIENCE

Maintain intelligent operation even when connectivity is limited.



SCALE

Build a connected network of emergency response nodes.



SMART DRONES. FASTER RESPONSE. SAVING LIVES.



MISSION SANJEEVANI

WHY MISSION SANJEEVANI CAN WIN

TECH SOCIETY SCALING WIN

A strong Specathon entry should be technically relevant, socially meaningful and scalable.

REAL PROBLEM

Targets the golden-hour emergency response gap.



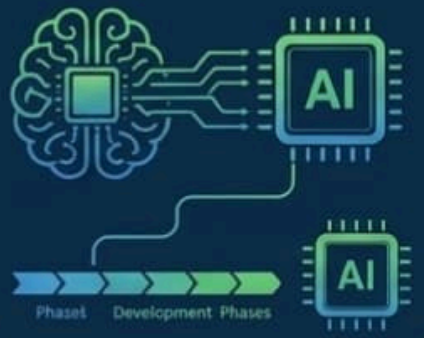
IoT CORE

Connectivity is central to the system—not an add-on.



AI ROADMAP

AI is enabled in the architecture, with modules being implemented and tested.



SCALABLE VISION

Solar-hybrid stations can expand the network into rural regions.



ONE PLATFORM • MULTIPLE TECHNOLOGIES • ONE HUMAN-CENTRIC MISSION

MISSION SANJEEVANI

An IoT Network That Turns Data Into Action

SENSE • CONNECT • ANALYZE • RESPOND • SAVE

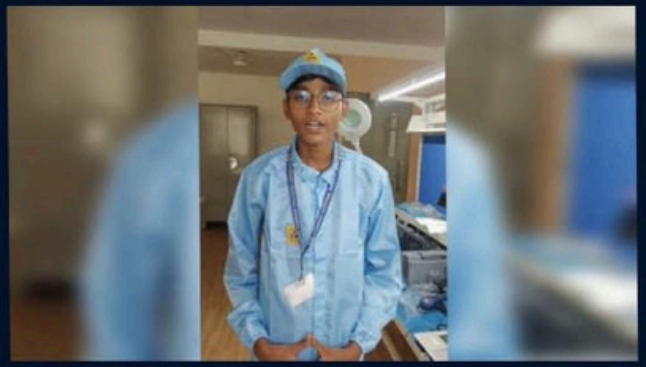
Smart Connectivity. Intelligent Response. Saving Lives.

"Every second counts. Every connected signal can make a difference."

THANK YOU | SPECATHON • IoT THEME

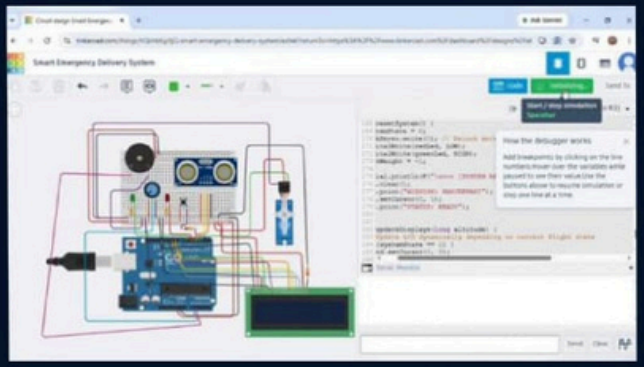
REAL PROTOTYPE EVIDENCE

Evidence captured from the uploaded project demonstration video.



TEAM PRESENTATION

Team member presenting the Mission Sanjeevani project.



IoT PROTOTYPE / SIMULATION

Circuit and control logic demonstration for the emergency-delivery system.



DEVELOPMENT EVIDENCE

Prototype development and testing evidence from the submitted video.

STATUS → Prototype development and IoT control logic demonstrated; AI modules are being implemented and tested as the next development stage.

WHY OUR IoT APPROACH IS CREDIBLE

We are building step-by-step: prototype first, AI validation next, scalable AIoT network ahead.

PROBLEM-DRIVEN

Addresses a real emergency-response and golden-hour challenge.



IoT-CENTRIC

Connectivity and sensing form the foundation of the system.



PROTOTYPE EVIDENCE

The submitted video demonstrates active project development and system logic.



AI ROADMAP

AI features are being implemented and tested—not presented as fully complete.



OUR PROMISE → Build, validate, connect, and scale technology that can help emergency response.